

MTE 204: Numerical Methods

1.0 Introduction

Duane Cronin

Department of Mechanical Engineering

University of Waterloo

(519) 888-4567 x32682

dscronin@uwaterloo.ca

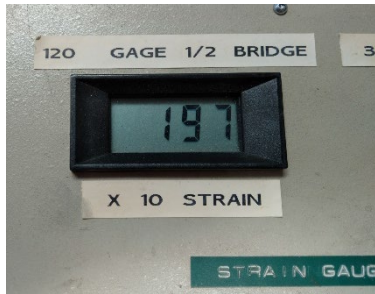
1.0 Introduction [Engineering Problem Solving]

- Engineering problem solving – two approaches

1. Observations, experiments, data

2. Fundamental laws, relationships -> expressed in the form of mathematical models

Strain gauge on a beam in bending



Equilibrium on a structure
(from Newton's 2nd Law)

$$\sum \vec{F} = 0 \quad \text{and} \quad \sum \vec{M} = 0$$

or

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum F_z = 0$$

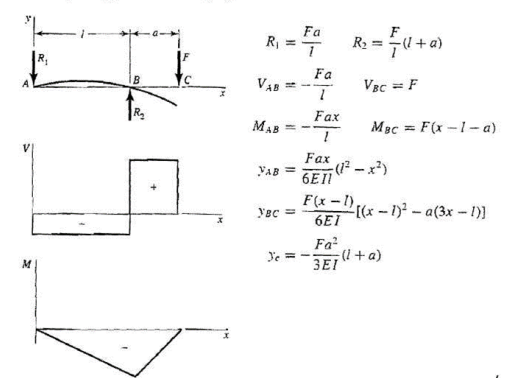
or

$$\sum M_x = 0$$

$$\sum M_y = 0$$

$$\sum M_z = 0$$

10 Simple supports—overhanging load



Beam Strain Calculation		
yc	-0.588 in	
l	1 in	
a	9.299 in	
E	1.03E+07 psi	
t	2.50E-01 in	
w	1.00E+00 in	
I	1.30E-03 in^4	
F	26.56482944 lbs	
x	2.329 in	
ybc	-0.023650966 in	
x	2.329 in	
Mbc	-211.7216907 in-lb	
stress	2.03E+04 psi	
strain	1.97E-03 (-)	
microstrain	1973 (ustrain)	
microstrain x10	197 (ustrain /10)	

1.0 Introduction [Mathematical Models]

- Divided into two categories
1. Analytical models – solve equation(s)
 2. Numerical models – approximation based on discretization

Numerical methods (e.g. Finite Element)

Implicit – solving an equation involving the current time and the later time simultaneously. (in implicit structural analysis, this is sometimes called the load step, rather than the time step).

$$\{F\} = [K]\{u\}$$

Force = Global Stiffness × Nodal displacements

Explicit – calculation of the state of a system at a later time from the current time (e.g. time stepping).

$$\{F_{External}\} = \{F_{Internal}\} + [C]\{\dot{D}\} + [M]\{\ddot{D}\}$$

Damping Mass x Accel.

- We solve the equation using the central difference method:

$$\{\dot{D}\} = \frac{1}{2\Delta t} [\{D\}_{n+1} - \{D\}_{n-1}] \text{ (Velocity)}$$

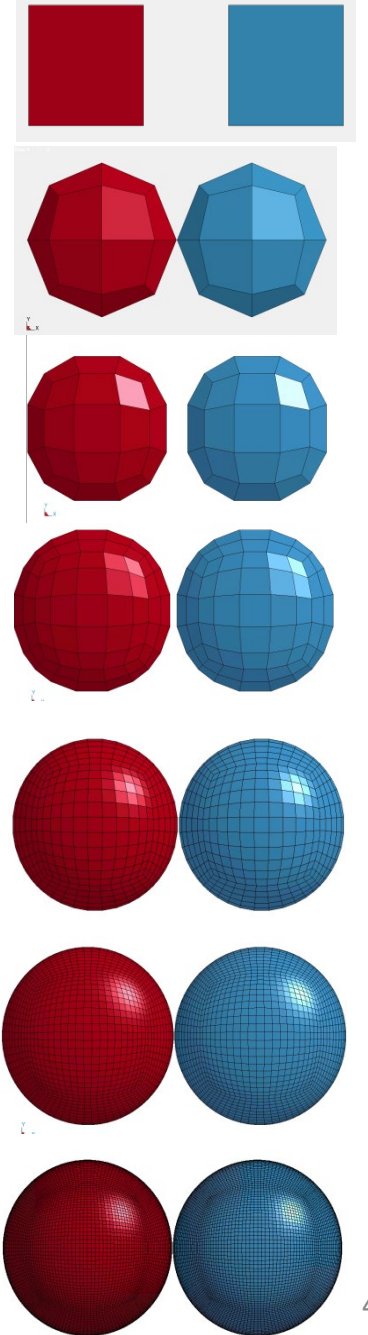
$$\{\ddot{D}\} = \frac{\Delta V}{\Delta t} = \frac{1}{\Delta t^2} [\{D\}_{n+1} - 2\{D\}_n + \{D\}_{n-1}] \text{ (Acceleration)}$$

- Note that the calculations depend on the appropriate selection of the time step (Δt).



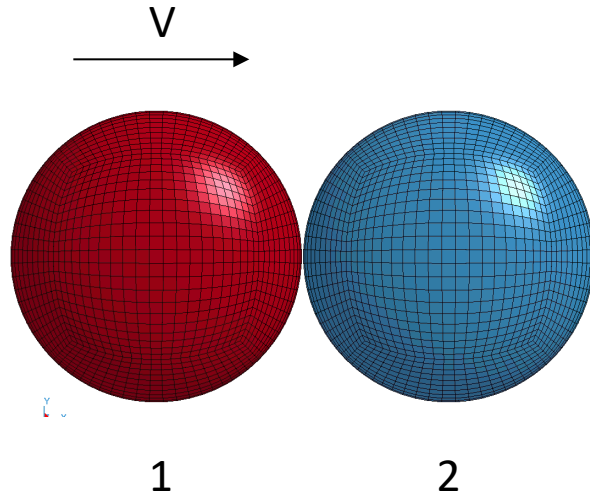
1.0 Introduction [Mathematical Model]

- Expresses essential features of physical system in mathematical terms
 - Divided into two categories
1. Analytical models – solve equation
 - “Exact” solution is achievable
 - Can be mathematically intense, challenging or impossible to solve
 2. Numerical models – approximation based on discretization
 - Discretization – breaking the problem up into smaller pieces (e.g. elements)
 - Solving simple (often algebraic) equations in small, subintervals
 - Allows us to address complex geometry (shape), nonlinear materials, complex boundary conditions (loading)
 - Sometimes the only way to solve complex problems



1.0 Introduction [Mathematical Model]

- Impact of two spheres (1-D problem) analytical solution
 - Rigid spheres ($r=0.1\text{m}$), conservation of momentum and energy



	initial		final	
m1	31.19 kg		31.19 kg	
v1	1.00 m/s		0.00 m/s	
E	15.59 J		0.00 J	
m2	31.19 kg		31.19 kg	
v2	0.00 m/s		1.00 m/s	
E	0.00 J		15.59 J	

$$V_{Af} = \frac{(m_A - m_B)}{(m_A + m_B)} V \quad V_{Bf} = \frac{2m_A}{(m_A + m_B)} V$$

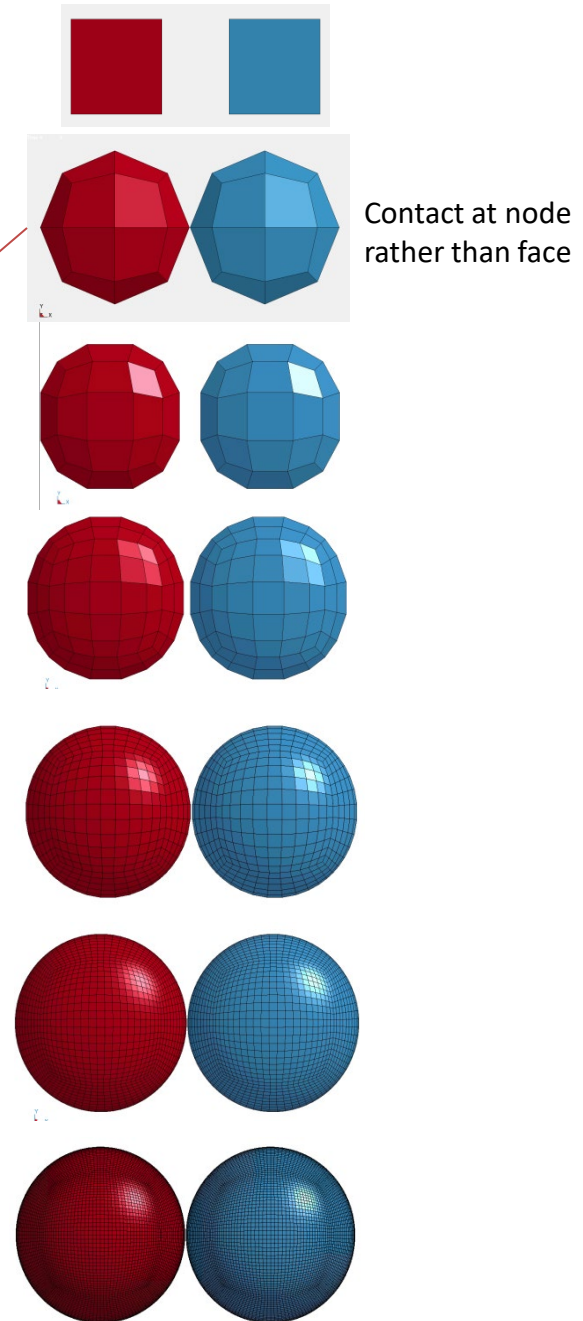
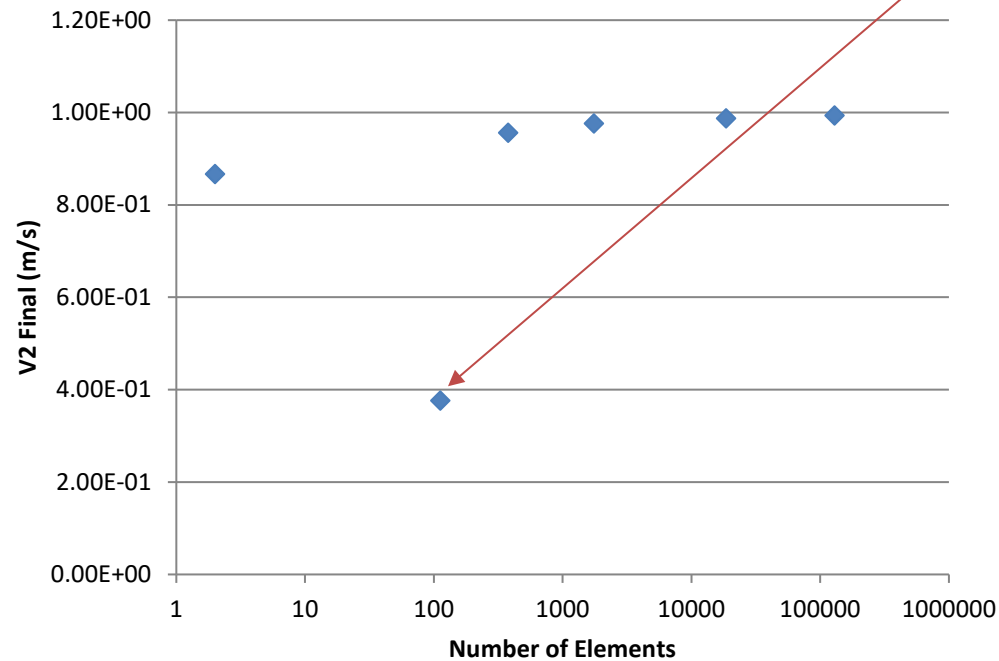
For $V=1\text{m/s}$, $V_{Af}=0.0$ and $V_{Bf}=1.0 \text{ m/s}$

Later, we will find that this solution is a good approximation for elastic spheres.

1.0 Introduction [Mathematical Model]

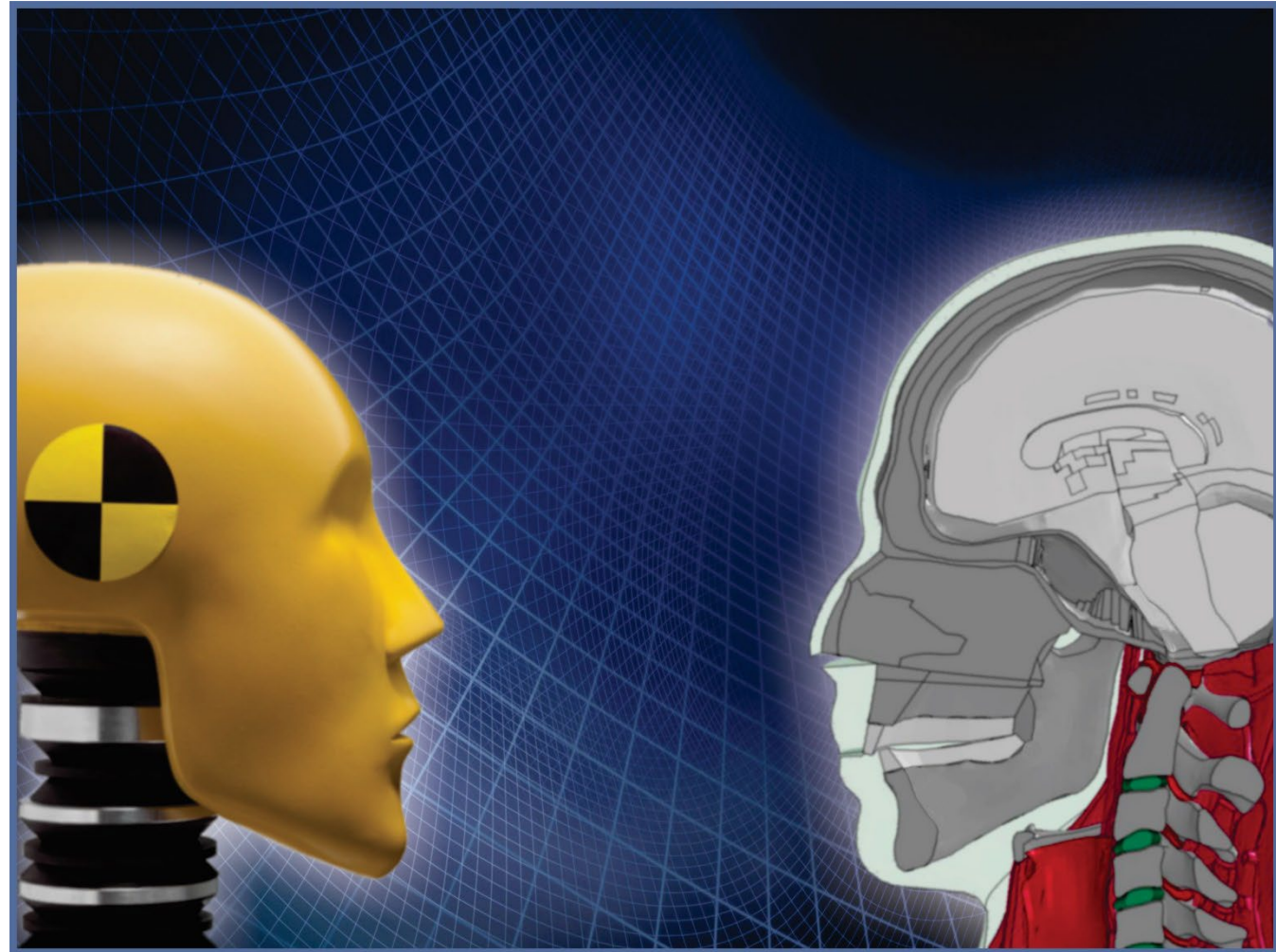
- Impact of two spheres (1-D problem)
numerical solution

Effect of mesh resolution on predicted velocity



1.0 Introduction

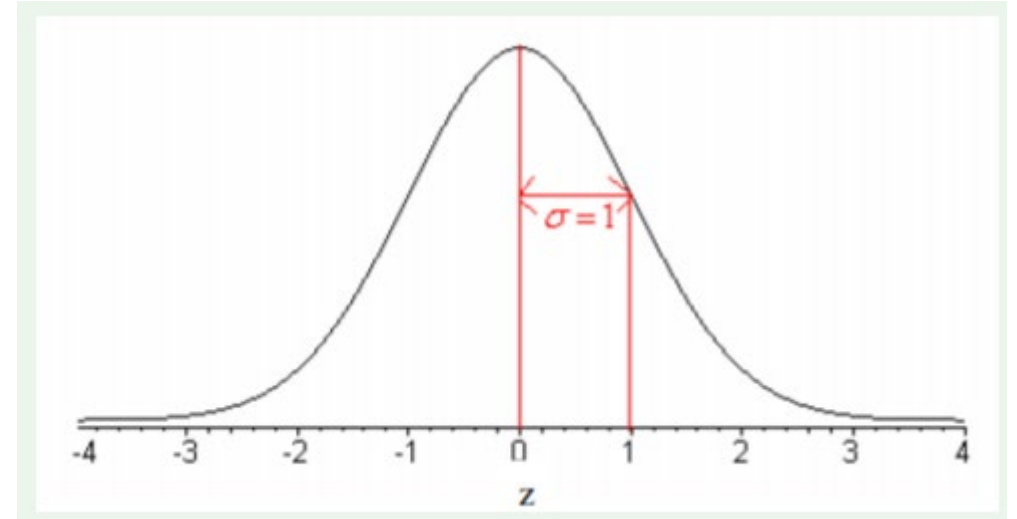
- Numerical methods and computational modeling
[background presentation – not on exam!]



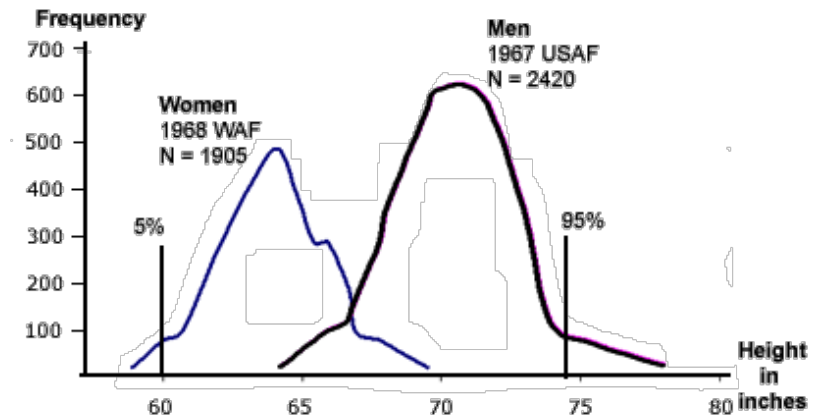
1.0 Introduction [Example]

- “Normally” distributed data, modeled by standard normal probability density function

- $$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



Anthropometrics
human (anthro) measurements (metric).



1.0 Introduction [Example]

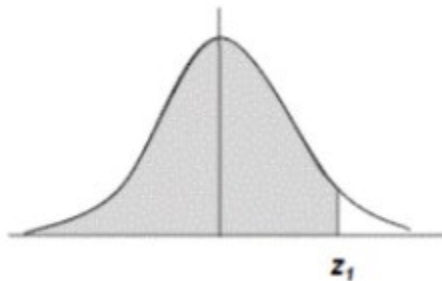
- “Normally” distributed data, modeled by standard normal probability density function

- $$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- If $\mu = 0, \sigma = 1$ find $P(0 < x < 1)$
- We can use the standard normal tables

- Or

$$z = \frac{x - \mu}{\sigma}$$

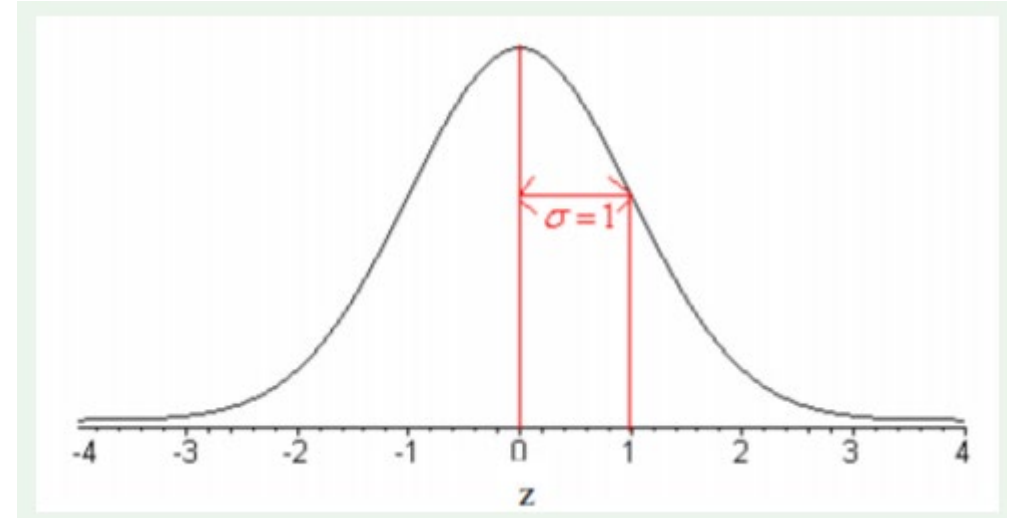


$$p(z \leq z_1) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z_1} e^{-\frac{1}{2}z^2} dz$$

$$= f(x) = \frac{1}{2} \cdot \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) \rightarrow \text{analytical solution, difficult}$$

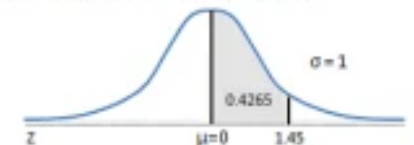
$$= 0.8413 - 0.5 = 0.3413 \rightarrow \text{numerical integration easy to use}$$

Interpretation: ~34% of the observations are within 1σ or $\pm 1\sigma$ contains ~68% of the observations



Areas Under the One-Tailed Standard Normal Curve

This table provides the area between the mean and some Z score.
For example, when Z score = 1.45 the area = 0.4265.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319



1.0 Introduction [Example 1.1 – A Parachutist, pg 14, 8th edition]

- A parachutist of mass 68.1 kg jumps out of a stationary hot air balloon.
- Use Equations 1.10 and 1.12 to compute the terminal velocity in freefall, prior to opening the parachute.
- The drag coefficient (c) is equal to 12.5 kg/s.

$$\vec{F} = m\vec{a}$$

$$\frac{d\vec{v}}{dt} = \frac{\vec{F}}{m}$$

$$\vec{F} = \vec{F}_{gravity} + \vec{F}_{drag}$$

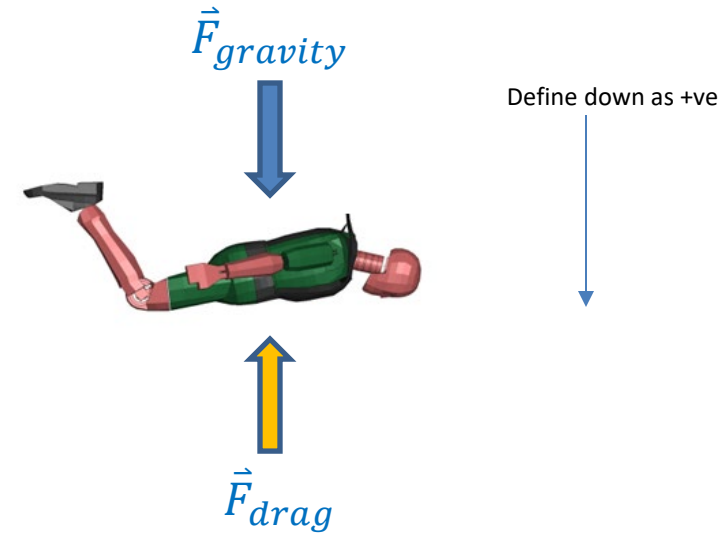
$$\vec{F}_{gravity} = m\vec{g} [down]$$

$$\vec{F}_{drag} = -c\vec{v} [up]$$

$$\frac{d\vec{v}}{dt} = \vec{g} - \frac{c}{m}\vec{v} \Rightarrow \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} = \vec{g} - \frac{c}{m}\vec{v}$$

- Analytical solution (Equation 1.10)
 - $v(t) = \frac{\vec{g}m}{c}(1 - e^{-(\frac{c}{m})t})$
- Numerical solution (Equation 1.12)
 - $v(t_{i+1}) = v(t_i) + \left[\vec{g} - \frac{c}{m}v(t_i)\right](t_{i+1} - t_i)$

... in freefall at constant velocity



Aside: kinematics resulting from parachute deployment



1.0 Introduction [Example 1.1 – A Parachutist]

Numerical solution:

- Using Microsoft Excel, make velocity versus time charts with different time intervals (Δt) of 0.1, 1, and 5 seconds, using Equation 1.12 from the textbook

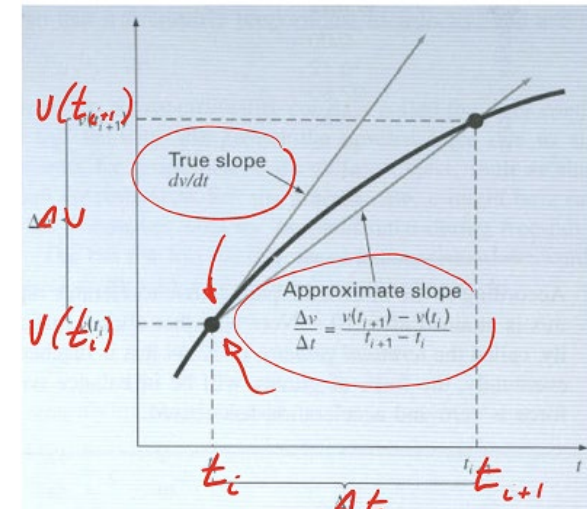
$$v(t_{i+1}) = v(t_i) + \left[\vec{g} - \frac{c}{m} v(t_i) \right] (t_{i+1} - t_i)$$

$$\Delta t = (t_{i+1} - t_i)$$

Analytical solution

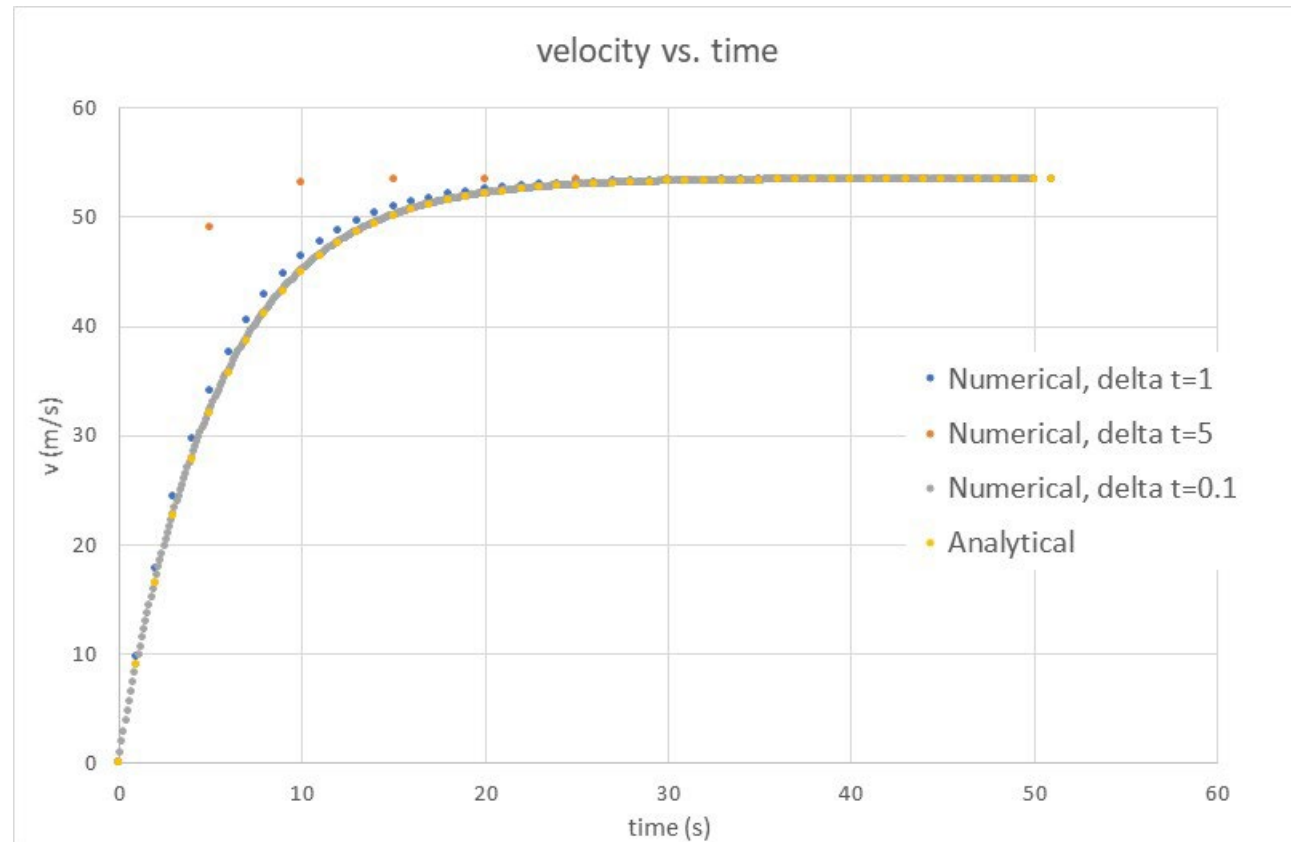
- Make another chart where the velocity is calculated using Equation 1.10 from the textbook. Plot for a time increment of 1s (note this is not the same as Δt above!).

Create a plot with all charts on one graph



1.0 Introduction [Example 1.1 – A Parachutist]

- Using Microsoft Excel, make velocity versus time charts with different time intervals (increase by 0.1, 1, and 5) with Equation 1.12 from the textbook
- Make another chart where the velocity is calculated using Equation 1.10 from the textbook. Plot for a time increment of 1s (note this is not the same as Δt).
- Create a plot with all charts on one graph



See provided spreadsheet for Example 1.1.

1.0 Introduction [Numerical Errors]

- Numerical errors arise from the use of approximations to represent exact mathematical operations and quantities.
 - True value = approximation + error
 - E_t = “true” error, exact value of error = true value - approximation

Error Definitions

- The true percent relative error when the true value is given

$$\varepsilon_t = \frac{\text{true error}}{\text{true value}} \cdot 100\%$$

- The approximate percent relative error when the true value is unavailable (in most actual situations)

$$\varepsilon_a = \frac{\text{approximate error}}{\text{approximate value}} \cdot 100\%$$

$$\varepsilon_a = \frac{\text{current approximation} - \text{previous approximation}}{\text{current approximation}} \cdot 100\%$$

- Used in iterative methods, where in each subsequent trial, the approximate value (hopefully) gets closer to the true value

- We usually use the absolute value of the error to avoid positive and negative values (in case we are summing error terms).
[see next]

1.0 Introduction [Numerical Errors]

Absolute error (this is in the units of the value x):

$$\varepsilon_{t\ Absolute} = |x_{True} - x_{Calculated}|$$

Relative error (this is normalized, nondimensional):

$$\varepsilon_{t\ Relative} = \left| \frac{x_{True} - x_{Calculated}}{x_{True}} \right| \cdot 100\%$$

- It is good practice to check for divide by zero, and to limit the number of iterations in a code if convergence isn't achieved.

For iterative solutions, we need a stopping criterion (i.e. how many iterations should we do?). The criterion is based on the difference between the new estimate of the value (x_{i+1}) and the current value (x_i). When this difference is less than a specified tolerance (TOL), we say the solution has converged.

Absolute error stopping criterion: $|x_{i+1} - x_i| < TOL$

Relative error stopping criterion: $\left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| < TOL$

However, in some cases the solution may be close to zero (i.e. possibility of divide by zero). In that case, absolute error, or a hybrid version may be needed:

$$\frac{|x_{i+1} - x_i|}{\text{Max}(|x_{i+1}|, \theta)} < TOL \text{ where } \theta > 0 \text{ is a problem-specific value}$$

1.0 Errors

$$\epsilon_a = \frac{\text{current approximation} - \text{previous approximation}}{\text{current approximation}} \cdot 100\%$$

- Example 1.2b – approximate error calculation
- For example, calculate **exp(0.5)**, using a Maclaurin series

$$\exp x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

- 1 term:

$$e^x = 1$$



- 2 terms:

$$e^x = 1 + x$$



$$\epsilon_a = \frac{1.5 - 1}{1.5} 100\% = 33.3\%$$

- How many terms do we need? Depends on the error we are willing to accept.

exp(0.5)=1.64872127070

We are accurate to

3 sig fig (1.65) at 4 terms

4 sig fig (1.649) at 6 terms

note - the 0.5 is not related to the exp on the left of the slide

Aside: Error criterion to ensure we are accurate to at least 'n' significant figures

$$\epsilon_s = (0.5 \times 10^{2-n})\%$$

For 3 significant figures:

$$\epsilon_s = (0.5 \times 10^{2-3})\% = 0.05\%$$

$\epsilon_a < \epsilon_s$ for 6 terms (below).

This ensures we are accurate to at least 3 significant figures.

** but, may occur for fewer terms

From Scarborough, 1966
[Section 3.3 in Chapra]

Terms	Result	ϵ_t (%)	ϵ_a (%)
1	1	39.3	
2	1.5	9.02	33.3
3	1.625	1.44	7.69
4	1.645833333	0.175	1.27
5	1.648437500	0.0172	0.158
6	1.648697917	0.00142	0.0158

1.0 Errors

- Calculate **exp(0.25)**, using a Maclaurin series expansion.

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

- True value exp(0.25)=1.284025417

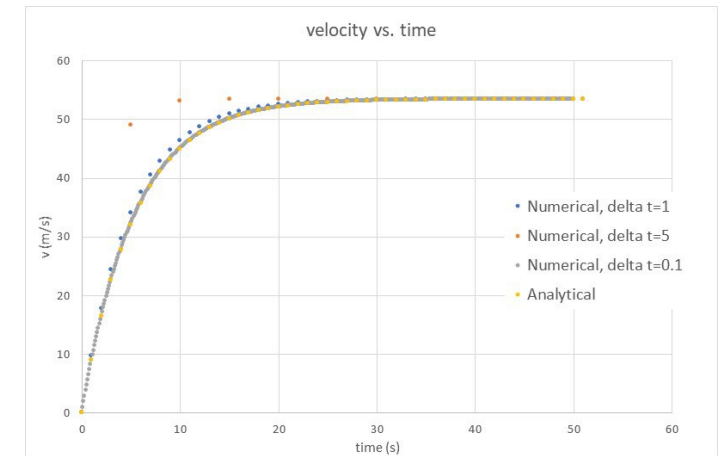
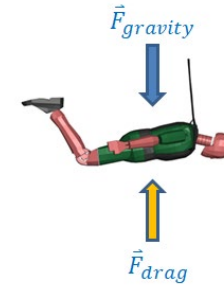
	Terms	exp(0.25)	ε_approx (%)	ε_true (%)
	1	1.0000000		22.119922
	2	1.2500000	20.000000	2.649902
	3	1.2812500	2.439024	0.216150
	4	1.2838542	0.202840	0.013337
	5	1.2840169	0.012676	0.000661
	6	1.2840251	0.000634	0.000027
	7	1.2840254	0.000026	0.000001

1.0 Errors

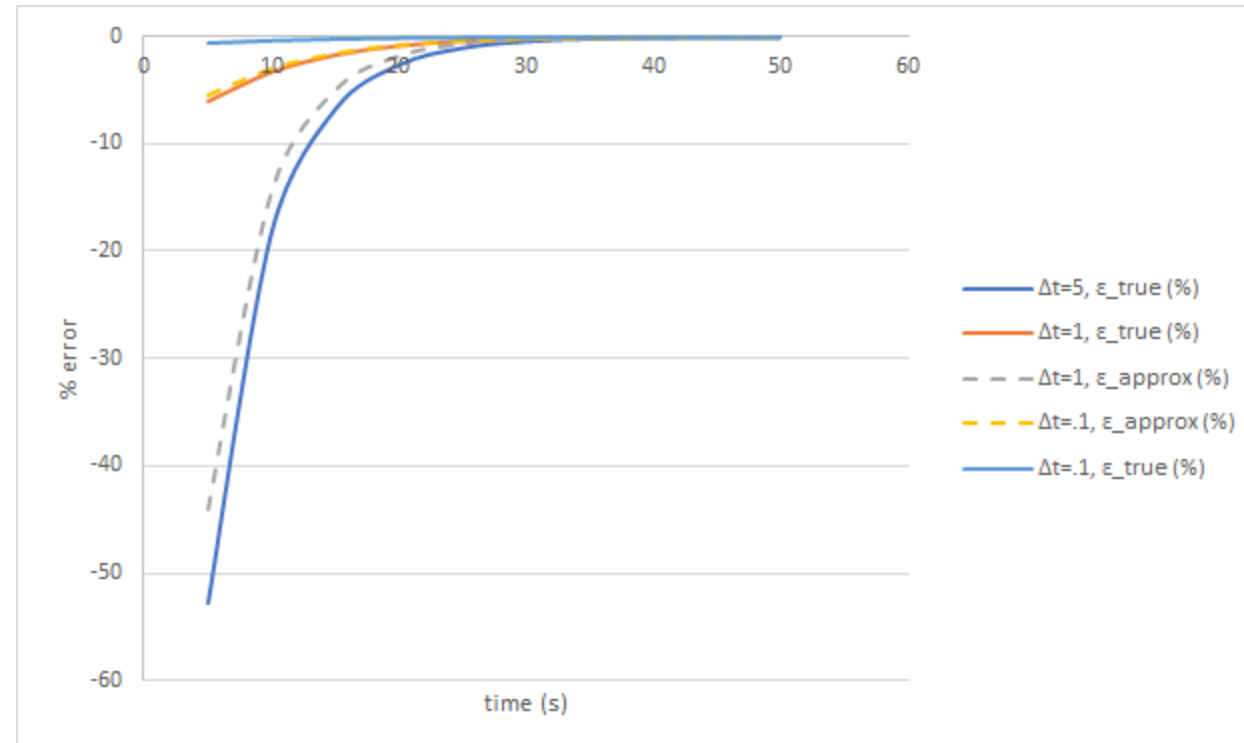
- Example 1.2 – true and approximate error calculation

$$\varepsilon_t = \frac{\text{true error}}{\text{true value}} \cdot 100\%$$

$$\varepsilon_a = \frac{\text{current approximation} - \text{previous approximation}}{\text{current approximation}} \cdot 100\%$$



t	$\Delta t=5$	$\varepsilon_{\text{true}} (\%)$	$\Delta t=1$	$\varepsilon_{\text{approx}} (\%)$	$\varepsilon_{\text{true}} (\%)$	$\Delta t=.1$	$\varepsilon_{\text{approx}} (\%)$	$\varepsilon_{\text{true}} (\%)$
0	0		0			0		
5	49.05	-52.8109	34.05637	-44.0259	-6.0996	32.27974	-5.5038	-0.5647
10	53.08348	-18.1762	46.4112	-14.3765	-3.3221	45.0631	-2.9916	-0.3210
15	53.41516	-6.7459	50.89323	-4.9553	-1.7060	50.12555	-1.5315	-0.1719
20	53.44244	-2.6067	52.5192	-1.7579	-0.8341	52.13036	-0.7459	-0.0876
25	53.44468	-1.0265	53.10907	-0.6319	-0.3921	52.92431	-0.3491	-0.0429
30	53.44486	-0.4076	53.32305	-0.2284	-0.1788	53.23872	-0.1584	-0.0203
35	53.44488	-0.1624	53.40068	-0.0828	-0.0796	53.36324	-0.0702	-0.0094
40	53.44488	-0.0648	53.42885	-0.0300	-0.0348	53.41255	-0.0305	-0.0043
45	53.44488	-0.0259	53.43906	-0.0109	-0.0150	53.43208	-0.0131	-0.0019
50	53.44488	-0.0103	53.44277	-0.0039	-0.0064	53.43981	-0.0055	-0.0008



1.0 Errors [Round-Off Errors]

- When numbers with limited significant figures are used to represent exact numbers
- $m \cdot b^e$
 - m=mantissa
 - b=base of the number system being used
 - e=exponent
 - e.g. $1.23 \cdot 10^4$ or 1.23×10^4 or 1.23E4
- When the mantissa has leading zero digits, it can be normalized by multiplying it by 10 and lowering the exponent by 1 to get an additional significant figure
 - e.g. $0.0123 \cdot 10^0$ to $0.1234 \cdot 10^{-1}$
- Computer round-off errors
 - Overflow error (when there are numbers that are too large or too small to be represented)
 - Quantizing errors (when irrational numbers cannot be represented e.g. Pi – ‘chop’ at a certain number of digits)
 - Error in the interval between numbers (when Δx increases as the number grows, the floating-point representation can keep its significant digits, but quantizing errors will be proportional to the magnitude of the number)
 - Machine Epsilon – ratio of Δx to x
 - The relative approximate error due to rounding numbers

Round-off occurs when we represent a real number in a floating-point system.

Machine epsilon relates to the relative approximate error (upper bound) due to rounding off. Think of it as the smallest positive floating-point number that, when added to 1.0, produces a result greater than 1.0.

80286 (286) 16-bit microprocessor



IEEE 754 - 2008	Common name	C++ data type	Base b	Precision p	Rounding machine epsilon ^[a] $b^{-(p-1)}/2$	Interval machine epsilon ^[b] $b^{-(p-1)}$
binary16	half precision	N/A	2	11 (one bit is implicit)	$2^{-11} \approx 4.88\text{e-}04$	$2^{-10} \approx 9.77\text{e-}04$
binary32	single precision	float	2	24 (one bit is implicit)	$2^{-24} \approx 5.96\text{e-}08$	$2^{-23} \approx 1.19\text{e-}07$
binary64	double precision	double	2	53 (one bit is implicit)	$2^{-53} \approx 1.11\text{e-}16$	$2^{-52} \approx 2.22\text{e-}16$

1.0 Introduction [Truncation Errors]

- From use of approximation in place of exact mathematical procedure

- Example

- Parachutist, terminal velocity

- $$\frac{dv}{dt} = \frac{\Delta v}{\Delta t}$$

Can be expressed in terms of Taylor series

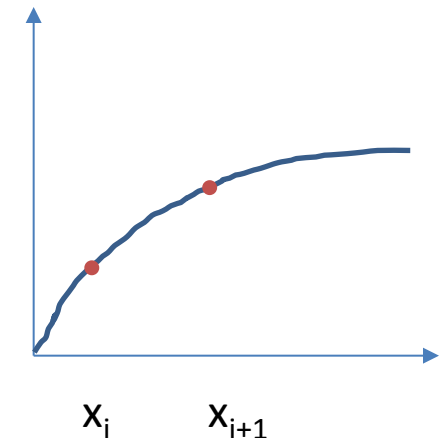
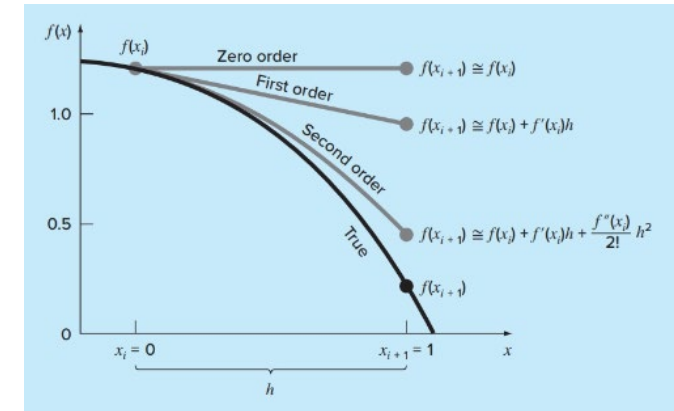
- If function f and its first $n+1$ derivatives are continuous on an interval of width h containing x , then the value of the function at $x + h$ is given by

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \dots$$

$$h = x_{i+1} - x_i$$

- Relates value of function at x_{i+1} to x_i

approximation of $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$ at $x = 1$



1.0 Introduction [Taylor Series – Rearranging Equation]

- $f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \dots$
- $h = x_{i+1} - x_i$

$$\frac{df(x_i)}{dx} = \frac{f(x_{i+1}) - f(x_i)}{h} - \frac{h}{2!} f''(x_i) - \frac{h^2}{3!} f'''(x_i) - \dots$$

Exact Approximation higher order terms (h.o.t.), truncation error

- Since $h \ll 1$ (assume, this is typically the case)
 - $\frac{h}{2!} f''(x_i) \gg \frac{h^2}{3!} f'''(x_i)$ and other h.o.t.
 - Truncation error of order h , first order
 - Truncation error is the difference between the true value of a function, and the value obtained from the numerical approximation.

1.0 Introduction [Control of Numerical Errors]

- True error (based on exact solution) not usually available, use approximate (estimated) error

$$\varepsilon_a = \frac{\text{approximate error}}{\text{approximate value}} \cdot 100\%$$

$$\varepsilon_a = \frac{\text{current approximation} - \text{previous approximation}}{\text{current approximation}} \cdot 100\%$$

- The true error can be difficult to find
- No systematic, general approach for error estimation for all problems – specific methods use different approaches
- A few guidelines
 - Avoid subtracting two nearly equal numbers
 - Do not add very small and very large numbers together
- Error control methods
 - Sensitivity analysis, such as grid refinement study or sensitivity to variations in input parameters
 - Examine limiting cases (e.g. upper/lower bounds)